

ROBOTIC ARM



**ESP32-Based IoT
Robotic Arm with
Web & Joystick
Control**

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Abstract

This project presents the design and development of an ESP32-based robotic arm capable of both manual and wireless control. The system integrates Internet of Things (IoT) technology to enable real-time communication between the robotic arm and a web-based dashboard using WebSockets. The robotic arm is constructed using four servo motors, which control the base rotation, arm movement, and gripper operation.

The ESP32 microcontroller acts as the central unit, processing inputs from both physical joystick modules and the web interface. The dual control system allows users to operate the robotic arm either locally through joysticks or remotely via a browser-based dashboard. The web interface includes features such as virtual joystick control, speed adjustment, and mode selection.

In addition to manual control, the system supports automation through manual path programming and playback functionality. Users can define sequences of movements with specific servo angles and speeds, enabling the robotic arm to perform repetitive tasks like pick-and-place operations efficiently.

To improve performance and stability, techniques such as dead zone filtering, smooth servo motion, and proper power management are implemented to reduce jitter and ensure reliable operation. The system provides a low-cost, flexible, and scalable solution for robotic automation.

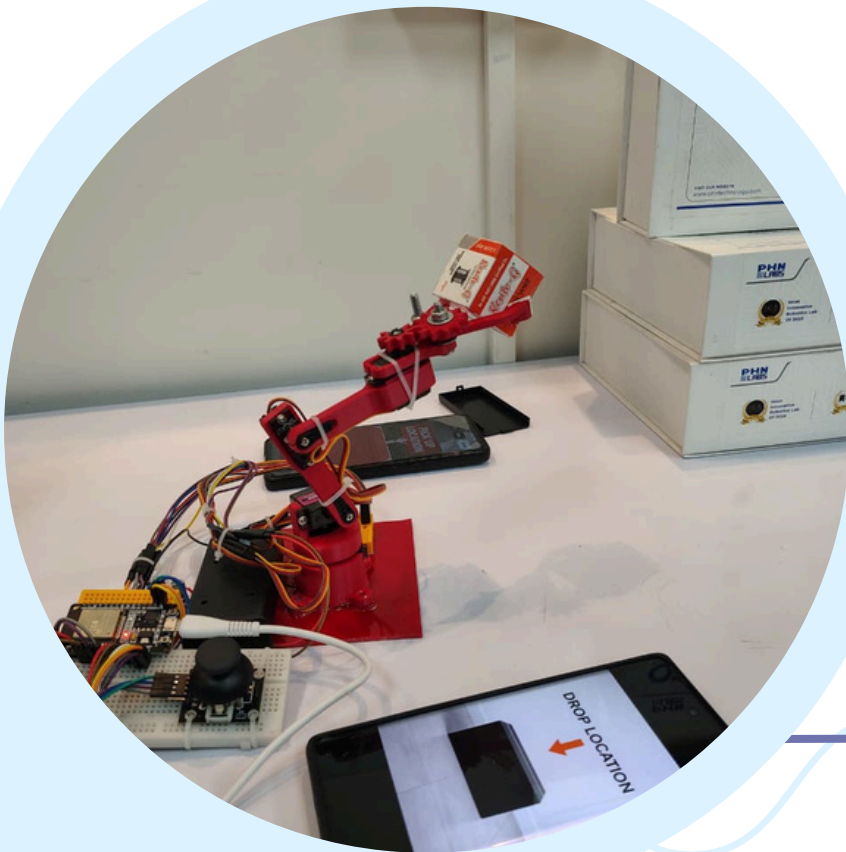
This project demonstrates the effective integration of embedded systems, wireless communication, and robotics, making it suitable for applications in industrial automation, education, and research.

About Project

The rapid growth of automation and robotics has increased the demand for intelligent and cost-effective robotic systems. This project focuses on designing a robotic arm that can be controlled both manually and remotely using modern IoT technologies.

The robotic arm is built using an ESP32 microcontroller, which provides built-in WiFi capabilities. It allows users to control the arm using physical joysticks as well as a web dashboard. The system supports advanced functionalities such as real-time control, programmable movement paths, and automation.

This project bridges the gap between traditional robotics and IoT-based smart systems, making it suitable for educational, industrial, and research applications.



Problem Statement And Objectives

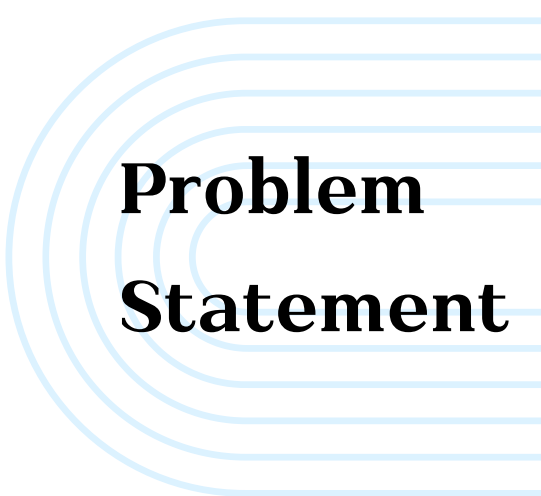


Objectives

- To design a robotic arm using ESP32
- To implement dual control (joystick + web dashboard)
- To enable wireless real-time control using WebSockets
- To implement manual path programming and playback
- To achieve smooth and stable servo motor operation

In many industries, repetitive tasks such as picking and placing objects require significant human effort and time. Existing robotic systems are often expensive and complex, making them inaccessible for small-scale applications and educational purposes.

There is a need for a low-cost, flexible, and easy-to-use robotic system that can perform such operations efficiently while being remotely controllable and programmable.

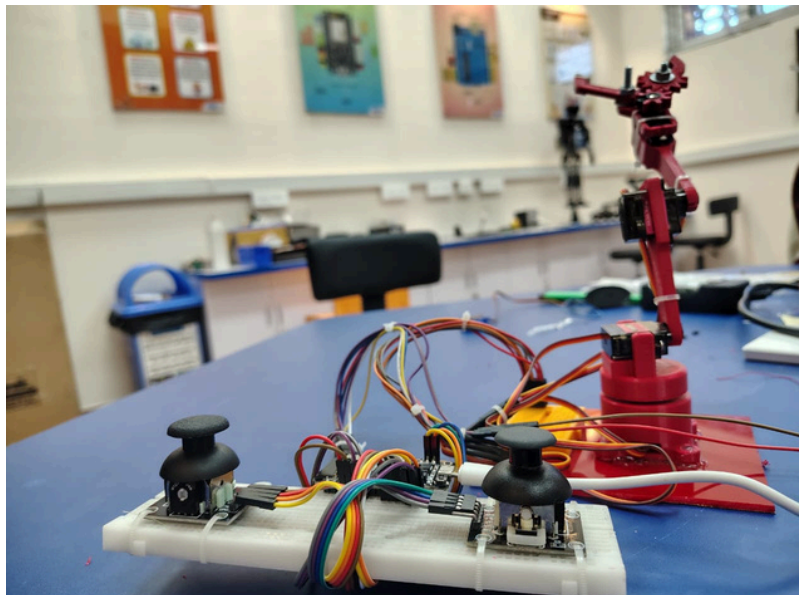


Problem Statement

System Overview

The system consists of input devices (joystick and web dashboard), a processing unit (ESP32), and output devices (servo motors).

- The joystick provides manual control
- The web dashboard provides remote control
- ESP32 processes all inputs and controls the servos
- Servo motors move the robotic arm accordingly



System Architecture

The system follows a simple Input–Process–Output model:

- Input: Joystick signals and web commands
- Process: ESP32 processes data and computes angles
- Output: Servo motors perform movements

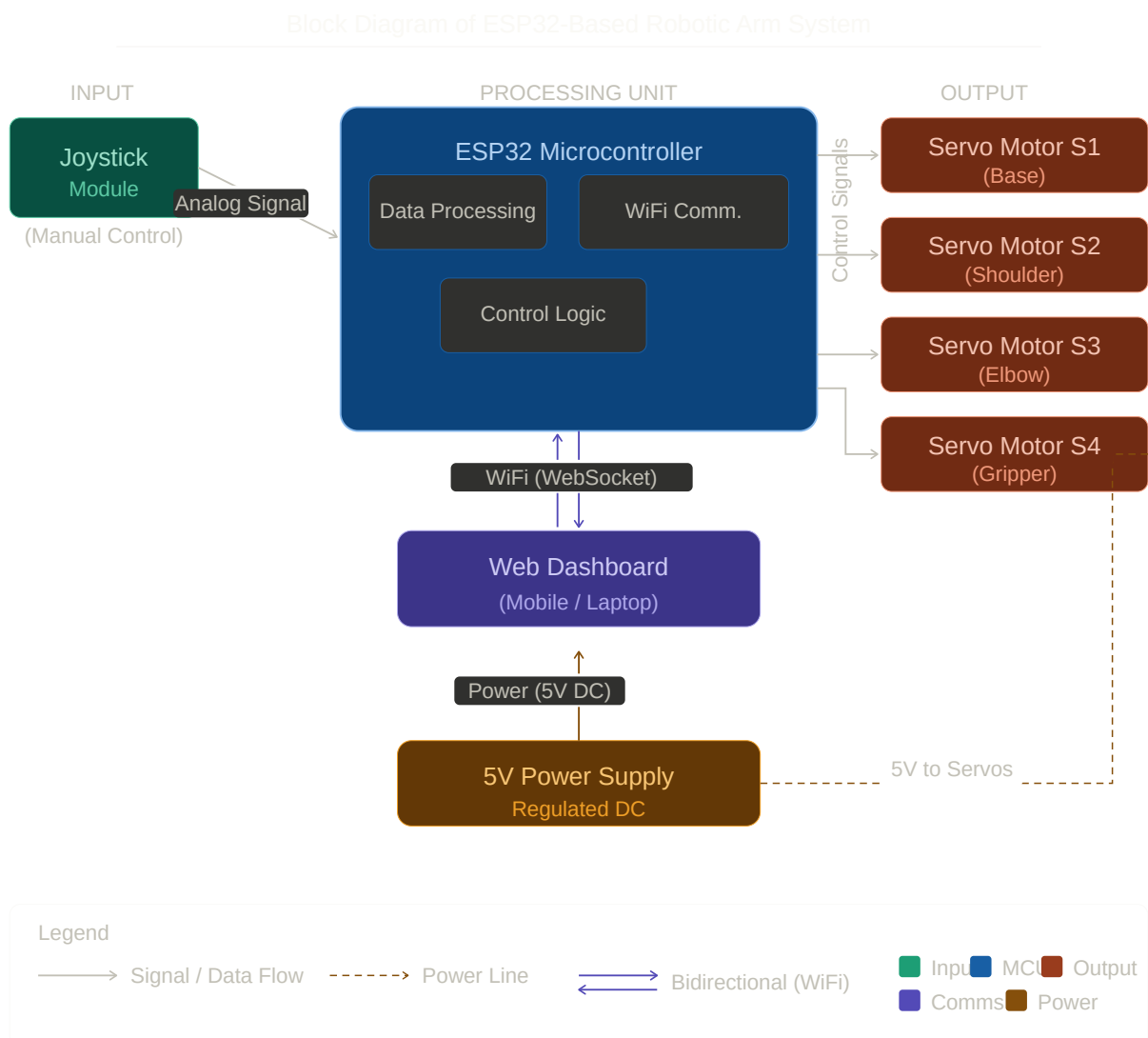
The ESP32 also acts as a server, enabling communication between the robotic arm and the web dashboard.

Hardware Components

Component	Quantity	Description
ESP32	1	Microcontroller with WiFi
Servo Motors	4	Used for arm movement
Joystick Module	2	Manual input device
Power Supply	1	External 5V supply
Jumper Wires	-	Connections

Circuit Description

The servo motors are connected to GPIO pins of the ESP32 and powered using an external 5V power supply. The joystick modules are connected to analog input pins. A common ground is maintained between all components to ensure stable operation. The ESP32 communicates with the web dashboard via WiFi.





Software Design

The software is developed using Arduino IDE. The following libraries are used:

- WiFi.h
- WebSocketsServer.h
- ArduinoJson.h
- ESP32Servo.h

The program reads inputs from the joystick and web interface, processes them, and controls the servo motors accordingly. It also handles communication and path execution.

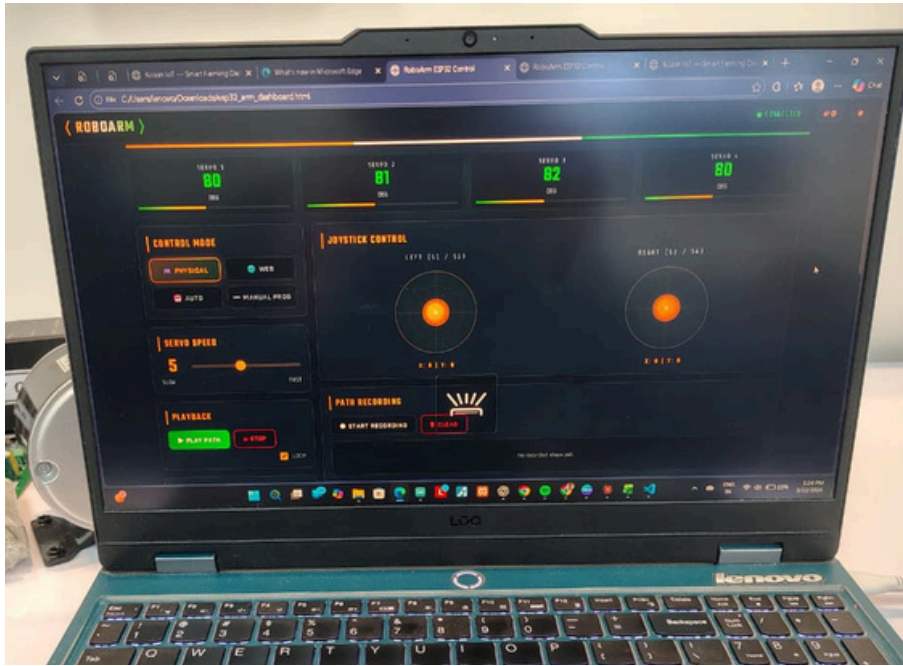


Web Dashboard

The web dashboard allows users to control the robotic arm remotely. It includes:

- Virtual joystick controls
- Speed control slider
- Mode selection
- Path recording and playback
- Manual path programming

Communication is established using WebSockets for real-time performance.



Working Principle

- User provides input through joystick or web dashboard
- ESP32 receives and processes the input
- Servo angles are calculated
- Servo motors move accordingly
- Feedback is sent back to the dashboard

Manual Path Programming

The system allows users to define movement steps manually using structured data. Each step includes servo angles and speed. The ESP32 executes these steps sequentially, enabling automation of tasks such as pick-and-place operations.

Results and Output

The robotic arm successfully performs:

- Manual control using joystick
- Remote control via web dashboard
- Automated pick-and-place operations
- Smooth and stable movement

Challenges Faced

- Servo motor shaking
- Power supply instability
- Noise in joystick signals
- Network delays

Solutions Implemented

- External power supply for servos
 - Dead zone filtering
 - Smooth movement algorithm
 - Optimized communication
-

Applications

- Industrial automation
- Pick-and-place systems
- Educational robotics
- Smart agriculture

Future Scope

- AI-based object detection
- Mobile app control
- Voice control system
- Cloud integration

Conclusion

The project successfully demonstrates a smart robotic arm controlled using ESP32. It integrates IoT with robotics to provide flexible, efficient, and cost-effective automation. The system is scalable and can be enhanced with advanced technologies in the future

